

S SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
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Section / Batch:	2025-2 nd year	Date:	24/07/2026

Code of Conduct

I, S Dharshini
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: S Dharshini

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Circular Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue+Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
    if (isdigit(expression[i])) push(expression[i] - '0');
    else {
        int op1 = pop();
        int op2 = pop();
        // perform operation: result = op1 + op2;
        push(result);
    }
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>
#define MAX 100
int stack[MAX];
int top = -1;

void push(int x) {
    if (top < MAX - 1) {
        stack[++top] = x;
    }
}

int pop() {
    if (top == -1) {
        return -1;
    }
    return stack[top--];
}

int evaluate postfix(char expression[]) {
    int i = 0;

    while (expression[i] != '\0') {
        if (isdigit(expression[i])) {
            push(expression[i] - '0');
        } else {
            int op1 = pop();
            int op2 = pop();
            int result = 0;

            switch (expression[i]) {
                case '+':
                    result = op1 + op2;
                    break;
                case '-':
                    result = op1 - op2;
                    break;
                case '*':
                    result = op1 * op2;
                    break;
                case '/':
                    result = op1 / op2;
                    break;
            }

            push(result);
        }
        i++;
    }

    return pop();
}

int main() {
    char expression[MAX];

    scanf("%s", expression);

    printf("%d\n", evaluate postfix(expression));

    return 0;
}
```

Submitted

Test Input

23+

Output

5

Pending manual review

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Fourth Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (l >= r) {
5         return -1;
6     }
7     int mid = l + (r - l) / 2;
8     if (arr[mid] == x) {
9         return mid;
10    }
11    if (arr[mid] > x) {
12        return binarySearch(arr, l, mid - 1, x);
13    }
14    return binarySearch(arr, mid + 1, r, x);
15 }
16
17 int main() {
18     int n, x, arr[100];
19     scanf("%d", &n);
20     for (int i = 0; i < n; i++) {
21         scanf("%d", &arr[i]);
22     }
23     scanf("%d", &x);
24     int result = binarySearch(arr, 0, n - 1, x);
25     printf("%d\n", result);
26     return 0;
27 }
```

Test Input

5
10
20
30

Output

-1

Pending manual review

Submitted

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?

```
PEEK(stack);  
return stack[top+1]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 void push(int x) {  
9     if (top < MAX - 1) {  
10         stack[++top] = x;  
11     }  
12 }  
13  
14 int peek() {  
15     if (top == -1) {  
16         printf("Underflow\n");  
17         return -1;  
18     }  
19     return stack[top];  
20 }  
21  
22 int main() {  
23     int n, x;  
24  
25     scanf("%d", &n);  
26  
27     for (int i = 0; i < n; i++) {  
28         scanf("%d", &x);  
29         push(x);  
30     }  
31  
32     x = peek();  
33     if (x != -1) {  
34         printf("%d\n", x);  
35     }  
36  
37     return 0;  
38 }
```

Submitted

Test Input

5
10
20
30
40

Output

50

Pending manual review

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Expression
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertfront operation.

```
void insertfront(int data) {
    if (front == 0) front = MAX - 1;
    else front = front - 1;
    deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int deque[MAX];
6 int front = -1, rear = -1;
7
8 void insertfront(int data) {
9     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
10         printf("Overflow\n");
11         return;
12     }
13
14     if (front == -1) {
15         front = rear = 0;
16     } else if (front == 0) {
17         front = MAX - 1;
18     } else {
19         front = front - 1;
20     }
21
22     deque[front] = data;
23 }
24
25 void display() {
26     if (front == -1) {
27         printf("Empty\n");
28         return;
29     }
30
31     int i = front;
32     while (1) {
33         printf("%d ", deque[i]);
34         if (i == rear) break;
35         i = (i + 1) % MAX;
36     }
37     printf("\n");
38 }
39
40 int main() {
41     int n, value;
42
43     scanf("%d", &n);
44     for (int i = 0; i < n; i++) {
45         scanf("%d", &value);
46         insertfront(value);
47     }
48
49     display();
50     return 0;
51 }
```

Submitted

Test Input

5
10
20
30
40

Output

40 30 20 10 10

Pending manual review

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Dequeue
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Expression
5 marks

11/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

Test Input

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 void enqueue(int data) {
9     if ((rear + 1) % MAX == front) {
10         printf("Queue full\n");
11         return;
12     }
13
14     if (front == -1) {
15         front = rear = 0;
16     } else {
17         rear = (rear + 1) % MAX;
18     }
19
20     queue[rear] = data;
21 }
22
23 int dequeue() {
24     if (front == -1) {
25         return -1;
26     }
27
28     int data = queue[front];
29
30     if (front == rear) {
31         front = rear = -1;
32     } else {
33         front = (front + 1) % MAX;
34     }
35
36     return data;
37 }
38
39 int main() {
40     int n, x, removed;
41
42     scanf("%d", &n);
43
44     for (int i = 0; i < n; i++) {
45         scanf("%d", &x);
46         enqueue(x);
47     }
48
49     removed = dequeue();
50
51     if (removed == -1) {
52         printf("Queue empty\n");
53     } else {
54         printf("%d\n", removed);
55     }
56
57     return 0;
58 }
```

5

10

20

30

...

Output

10

🔴 Pending manual review

📌 Submitted

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Dequeue
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stacks - Prefix Suffixes
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 void enqueue(int data) {
9     if (rear == MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13
14     if (front == -1) {
15         front = 0;
16     }
17
18     queue[++rear] = data;
19 }
20
21 int dequeue() {
22     if (front == -1 || front > rear) {
23         printf("Underflow\n");
24         return -1;
25     }
26
27     return queue[front++];
28 }
29
30 int main() {
31     int n, value;
32
33     scanf("%d", &n);
34
35     for (int i = 0; i < n; i++) {
36         scanf("%d", &value);
37         enqueue(value);
38     }
39
40     while (front <= rear) {
41         printf("%d ", dequeue());
42     }
43
44     return 0;
45 }
```

Test Input

5
10
20
30

Output

10 20 30 40 50

Pending manual review

Submitted

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Infix to Postfix
5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

⚠ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct node {
5     int data;
6     struct node *prev;
7     struct node *next;
8 } node;
9
10 node *createNode(int data) {
11     node *newNode = (node *)malloc(sizeof(node));
12     newNode->data = data;
13     newNode->prev = NULL;
14     newNode->next = NULL;
15     return newNode;
16 }
17
18 void insertend(node **head, int data) {
19     node *newNode = createNode(data);
20
21     if (*head == NULL) {
22         *head = newNode;
23         return;
24     }
25
26     node *temp = *head;
27     while (temp->next != NULL) {
28         temp = temp->next;
29     }
30
31     temp->next = newNode;
32     newNode->prev = temp;
33 }
34
35 void deleteNode(node **head, node *del) {
36     if (*head == NULL || del == NULL) {
37         return;
38     }
39
40     if (*head == del) {
41         *head = del->next;
42     }
43
44     if (del->prev != NULL) {
45         del->prev->next = del->next;
46     }
47
48     if (del->next != NULL) {
49         del->next->prev = del->prev;
50     }
51
52     free(del);
53 }
54
55 void display(node *head) {
56     node *temp = head;
57     while (temp != NULL) {
58         printf("%d ", temp->data);
59         temp = temp->next;
60     }
61     printf("\n");
62 }
63
64 int main() {
65     node *head = NULL;
66
67     insertend(&head, 10);
68     insertend(&head, 20);
69     insertend(&head, 30);
70     insertend(&head, 40);
71
72     deleteNode(&head, head->next); // Delete 10
73
74     display(head);
75
76     return 0;
77 }
```

Submitted

Test Input

```
4
10 20 30 40
```

Output

```
10 20 40
```

Pending manual review

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked list
5 marks

Q9

Linked list
5 marks

Q10

Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front + 1]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int Q[MAX];  
6 int front = -1, rear = -1;  
7  
8 int dequeue() {  
9     if (front == -1 || front > rear) {  
10        printf("Underflow\n");  
11        return -1;  
12    }  
13  
14    int item = Q[front];  
15    front++;  
16  
17    if (front > rear) { // queue becomes empty  
18        front = rear = -1;  
19    }  
20  
21    return item;  
22 }  
23  
24 int main() {  
25     int n, i, x;  
26  
27     scanf("%d", &n);  
28  
29     if (n <= 0) {  
30         printf("Underflow\n");  
31         return 0;  
32     }  
33  
34     front = 0;  
35     rear = n - 1;  
36  
37     for (i = 0; i < n; i++) {  
38         scanf("%d", &Q[i]);  
39     }  
40  
41     x = dequeue();  
42  
43     if (x != -1) {  
44         printf("%d\n", x);  
45     }  
46  
47     return 0;  
48 }
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Stack - ordered List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q6 Stack

5 marks

Identify the errors in the following snippet.

```

if(ch == '{') {
    while(stack[top] != '}') {
        postfix[j++] = pop();
    }
    pop();
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4
5 #define MAX 100
6
7 char stack[MAX];
8 int top = -1;
9
10 void push(char c) {
11     stack[++top] = c;
12 }
13
14 char pop() {
15     return stack[top--];
16 }
17
18 char peek() {
19     return stack[top];
20 }
21
22 int precedence(char c) {
23     if (c == '^') return 3;
24     if (c == '*' || c == '/') return 2;
25     if (c == '+' || c == '-') return 1;
26     return 0;
27 }
28
29 void infixtopostfix(char infix[], char postfix[]) {
30     int i = 0, j = 0;
31     char ch;
32
33     while ((ch = infix[i]) != '\0') {
34         if (isalnum(ch)) {
35             postfix[j++] = ch;
36         } else if (ch == '(') {
37             push(ch);
38         } else if (ch == ')') {
39             while (top != -1 && peek() != '(') {
40                 postfix[j++] = pop();
41             }
42             if (top != -1 && peek() == '(') {
43                 pop();
44             }
45         } else {
46             while (top != -1 && precedence(peek()) >= precedence(ch)) {
47                 postfix[j++] = pop();
48             }
49             push(ch);
50         }
51         i++;
52     }
53
54     while (top != -1) {
55         postfix[j++] = pop();
56     }
57
58     postfix[j] = '\0';
59 }
60
61 int main() {
62     char infix[MAX], postfix[MAX];
63
64     scanf("%s", infix);
65
66     infixtopostfix(infix, postfix);
67     printf("%s", postfix);
68
69     return 0;
70 }

```

Test Input

abc*c

Output

abc*c

Pending manual review

Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

What is the error in the below code?

```
new->next = head;
head = new;
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 struct Node *head = NULL;
10
11 void insertAtBeginning(int value) {
12     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
13     if (newNode == NULL) {
14         printf("Memory allocation failed\n");
15         return;
16     }
17
18     newNode->data = value;
19     newNode->next = head;
20     head = newNode;
21 }
22
23 void display() {
24     struct Node *temp = head;
25     while (temp != NULL) {
26         printf("%d ", temp->data);
27         temp = temp->next;
28     }
29     printf("\n");
30 }
31
32 int main() {
33     int n, value;
34
35     scanf("%d", &n);
36     for (int i = 0; i < n; i++) {
37         scanf("%d", &value);
38         insertAtBeginning(value);
39     }
40
41     display();
42     return 0;
43 }
```

Test Input

```
5
10 20 30 40 50
```

Output

```
50 40 30 20 10
```

⏸ Pending manual review

📌 Submitted

QUESTIONS

Q1

Binary Search

5 marks



Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Single linked list

5 marks



Q7

Stack

5 marks



Q8

Linked list

5 marks



Q9

Linked list

5 marks



Q10

Queue

5 marks



10/30 answered

Q7 Stack

5 marks

Point out the error:

```
(POP)stack);  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define max 100  
3 int stack[max];  
4 int top = -1;  
5 void push(int item) {  
6     if (top == max - 1) {  
7         printf("Overflow\n");  
8     } else {  
9         stack[++top] = item;  
10    }  
11 }  
12  
13 int pop() {  
14     if (top == -1) {  
15         printf("Underflow\n");  
16         return -1;  
17     } else {  
18         return stack[top--];  
19     }  
20 }  
21  
22 int main() {  
23     int choice, x, item;  
24  
25     while (scanf("%d", &choice) != 1) {  
26         if (choice == 1) {  
27             if (scanf("%d", &x) != 1) break;  
28             push(x);  
29         } else if (choice == 2) {  
30             item = pop();  
31             if (item != -1) {  
32                 printf("%d\n", item);  
33             }  
34         } else if (choice == 3) {  
35             break;  
36         }  
37     }  
38     return 0;  
39 }  
40 }
```

Submitted

Test Input

```
1  
23  
1  
4
```

Output

```
23  
4  
23
```

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Referencing Symbolic
5 marks

Q6

Singly linked list
5 marks

Q7

Stack
5 marks

Q8

Linked list
5 marks

Q9

Linked list
5 marks

Q10

Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head;
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

🔍 **Debugging Task:** first find the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node* createnode(int data) {
10     struct node *newnode = (struct node*)malloc(sizeof(struct node));
11     newnode->data = data;
12     newnode->next = NULL;
13     return newnode;
14 }
15
16 void append(struct node **head, int data) {
17     struct node *newnode = createnode(data);
18
19     if (*head == NULL) {
20         *head = newnode;
21         return;
22     }
23
24     struct node *temp = *head;
25     while (temp->next != NULL) {
26         temp = temp->next;
27     }
28     temp->next = newnode;
29 }
30
31 void display(struct node *head) {
32     struct node *temp = head;
33
34     while (temp != NULL) {
35         printf("%d ", temp->data);
36         temp = temp->next;
37     }
38     printf("\n");
39 }
40
41 int main() {
42     struct node *head = NULL;
43
44     append(&head, 10);
45     append(&head, 20);
46     append(&head, 30);
47     append(&head, 40);
48
49     printf("linked list: ");
50     display(head);
51
52     return 0;
53 }
```

Test Input

stdin input (optional)

Output

linked list: 10 20 30 40

🔄 Pending manual review

📌 Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly linked list
5 marks

Q7

Stack
5 marks

Q8

Linked list
5 marks

Q9

Linked list
5 marks

Q10

Queue
5 marks

11/10 answered

Q5: Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```

if(ch == '[' && stack[top] == '[') {
    ch == '[' && stack[top] == '[') {
        pop();
    } else {
        return 0;
    }
}

```

🔍 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX 1000
5
6 int isbalanced(const char *expr) {
7     char stack[MAX];
8     int top = -1;
9
10    for (int i = 0; expr[i] != '\0'; i++) {
11        char ch = expr[i];
12
13        if (ch == '[' || ch == '{' || ch == '(') {
14            if (top < MAX - 1) {
15                stack[++top] = ch;
16            } else {
17                return 0;
18            }
19        }
20        else if (ch == ']' || ch == '}' || ch == ')') {
21            if (top < -1) {
22                return 0;
23            }
24
25            if ((ch == ']' && stack[top] == '[') ||
26                (ch == '}' && stack[top] == '{') ||
27                (ch == ')' && stack[top] == '(')) {
28                top--;
29            } else {
30                return 0;
31            }
32        }
33    }
34
35    return (top == -1);
36 }
37
38 int main() {
39     char expr[MAX];
40
41     if (fgets(expr, sizeof(expr), stdin) == NULL) {
42         return 0;
43     }
44
45     expr[strcspn(expr, "\n")] = '\0';
46
47     if (isbalanced(expr)) {
48         printf("1");
49     } else {
50         printf("0");
51     }
52
53     return 0;
54 }

```

Submitted

Test Input

{(())}

Output

1

Pending manual review

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

5/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 int stack[5];  
3 int top;  
4  
5 void init() {  
6     top = -1;  
7 }  
8  
9 int main(void) {  
10     init();  
11     printf("Stack initialised\n");  
12     printf("Top = %d\n", top);  
13     return 0;  
14 }
```

Test Input

stdin input (optional)

Output

```
Stack initialised  
Top = -1
```

Pending manual review

Submitted

QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack - Balancing Symbols
5 marks



Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

5/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[5];
3 int top = -1;
4 int pop() {
5     if (top == -1) {
6         printf("Underflow\n");
7         return -1;
8     }
9     return stack[top--];
10 }
11
12 void push(int val) {
13     if (top == 4) {
14         printf("Overflow\n");
15     } else {
16         stack[++top] = val;
17         printf("Pushed %d\n", val);
18     }
19 }
20
21 int main() {
22     push(10);
23     push(20);
24     printf("Deleted element: %d\n", pop());
25     return 0;
26 }
```

Submitted

Test Input

stdin input (optional)

Output

```
Pushed 10
Pushed 20
Deleted element: 20
```

Pending manual review

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack : Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

5/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int binarysearch(int arr[],int n,int key){
3     int low=0,high=n-1;
4     while(low<=high){
5         int mid=(low+high)/2;
6         if(arr[mid]==key)
7             return mid;
8         else if(arr[mid]<key)
9             low=mid+1;
10        else
11            high=mid-1;
12    }
13    return -1;
14 }
15 int main(){
16     int arr[]={2,4,6,8,12,15};
17     int n=sizeof(arr)/sizeof(arr[0]);
18     int key;
19     printf("enter element to search:");
20     scanf("%d",&key);
21     int result=binarysearch(arr,n,key);
22     if(result==1)
23         printf("element not found");
24     else
25         printf("element is found at index%d",result);
26     return 0;
27 }
```

Submitted

Test Input

8

Output

enter element to search:element is
found at index3

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